

AlphaGlass and the End of Backtest Theater

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Interpretability is becoming the only honest measure of out-of-sample edge, and the institutional question is which managers can produce it under stress.

Interpretability has graduated from compliance artifact to performance constraint

Sebastian Bell, Ali Kakhbod, Martin Lettau, and Abdolreza Nazemi released [AlphaGlass](#) through NBER earlier in the week, and the framing deserves attention from anyone allocating to systematic equity. The paper builds a characteristic-based portfolio model that is forced to expose, at every step, which firm characteristics and which interactions are driving each position. The discipline is a structural constraint that eliminates the overfitting black-box optimizers routinely hide behind in-sample

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AlphaGlass restricts the functional form so that every contribution to expected return can be traced back to a characteristic or a low-order interaction, and the out-of-sample performance holds up across regime changes precisely because the model cannot lean on spurious high-order combinations. The cost in expressiveness is real. The gain in robustness is larger.

For institutional allocation, the question has shifted from whether they can produce characteristic shocks to whether they can produce characteristic shocks of 2023 with decomposable drivers that are running.

The LLM stock checklist dissection

Elsewhere, an [arXiv](#) hedge-fund perspective

papers overstate live signal quality, and the catalog reads less like a critique of the literature and less like a primer than a stress test that every allocator should be running on external managers claiming LLM-driven alpha.

Three of the failure modes deserve to be named directly. First, illiquidity premia: most published LLM strategies post their best results in small-cap and micro-cap names where the transaction-cost assumptions implicit in the backtest would erase the signal in production. Second, data leakage: pretrained models have absorbed financial news earnings transcripts and analyst commentary whose publication dates postdate the



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with the news-release cadence rather than the trading cadence, which means the signal cannot be harvested at the stated frequency.

The implication for allocators is concrete. A manager who pitches LLM-driven alpha and cannot, in a diligence meeting, walk through how their pipeline addresses each of these three frictions with specific protocols is selling backtest theater. The honest managers in this space already know the checklist. They have already paid for the lookahead audits, already restricted their universe to liquid names where the signal survives transaction costs, and already separated their pretraining cutoff from their evaluation window with documented care. The dishonest ones will hand-wave.

SIGNAL

Goldman is betting that deployment speed beats organizational clarity, and publicly committing to the trade

Bloomberg [reported](#) John Waldron describing Goldman as a “human assembly line” facing automation, while explicitly refusing to forecast the headcount outcome. The signal is in the refusal. Waldron is telling competitors and clients that Goldman will restructure faster than it can model the end state, because in his read the first-move advantage in agent-scaled banking outweighs the reputational and operational cost of announcing job cuts before the org chart is known. Allocators with bank exposure should be modeling which competitors can credibly match that posture and which ones hesitate. The hesitation is where the margin gap opens.

The Agent Lab result that matters is the regression weight rather than the Brier score

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meaning the value comes from orthogonality rather than dominance. The right diligence question for an AI-assisted macro tool is whether the model's errors are independent enough of the market's errors to improve a blended estimate. Most tools have seen fail that test cleanly.

The AI roadmap for institutional finance is visible in job postings four months before it is visible in product launches

[AI Might Be Wrong](#) used Mercor job-board data to argue that the workflows being automated next are pre-announced through hiring patterns. For allocators, the use case is direct. The external managers and bank counterparties who are converting junior analyst, offshore execution, and middle-office work into AI-enabled product lines before competitors do will compound a margin advantage that sell-side estimates are not yet pricing. I am running this scan now on the top fifteen managers in our diligence pipeline, and the dispersion in posting patterns is wider than any other operational signal I track.

EVIDENCE

Multi-agent LLM systems earn their keep as universe filter, not stock pickers

An arXiv [paper](#) on signal-or-noise in multi-agent LLM stock recommendations reports a 0.489 ICIR for a system that rotates agent weights by macro regime. That number is unremarkable as a standalone alpha signal and genuinely useful as a dynamic universe filter that reshapes the opportunity set a fundamental or systematic manager evaluates. The institutional adoption case sits upstream of alpha generation.

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Individually rational LLM agents aggregate into bubble dynamics that mirror human experimental markets

This [paper](#) on dissecting AI trading shows that LLM agents, each acting on locally sensible reasoning, produce price paths in simulation that reproduce the bubble-and-crash patterns documented in Smith-style experimental asset markets. The authors further demonstrate that prompt engineering amplifies or suppresses the bubble at will. The simulation caveat matters and the implication still lands: for allocators building AI-driven trading infrastructure, prompt governance is a market-stability control rather than a UX preference. Whoever writes the prompts controls the bubble mechanics.

Stronger reasoning in LLM agents amplifies speculative coordination rather than dampening it

An SSRN [paper](#) on AI bubbles with large language models pushes the same finding further. Agents equipped with stronger chain-of-thought reasoning still coordinate speculative bubble equilibria, and the chain-of-thought traces reveal framing-sensitive belief formation rather than genuine price discovery. The result inverts the common assumption that more capable models make markets more efficient. Institutional allocators treating model sophistication as a safeguard against bubble participation are mispricing the risk in the opposite direction from the one they think.

Q-learning and LLM architectures produce structurally opposite systemic failure modes

The ECB working [paper](#) Ex Machina models AI adoption in financial markets at the architecture level and finds that Q-learning agents over-coordinate, amplifying

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regulators who treat AI architecture as an implementation detail are mispricing the systemic risk they are running. The architecture choice is the risk choice.

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